

SIMULATED SATELLITE-INDEX CROP CLASSIFICATION & GROWTH-STAGE-AWARE WATER-DEFICIT ADVISORY

Track: Software — Smart Agriculture & Precision Farming / Climate Action & Environmental Sustainability (self-simulated data only)

Problem Statement

Command-area irrigation planning depends on knowing what crop is growing where, and how water-stressed it is, at a spatial and temporal resolution finer than most publicly distributed satellite products actually deliver — real optical satellites revisit a given field only every few days, and cloud cover during the growing season creates long observation gaps. Sourcing, licensing, and validating live satellite or national irrigation datasets is not achievable inside a hackathon build window, so this problem statement is re-based on a team-built simulation: build and honestly disclose your own simulator of what a satellite vegetation/moisture-index feed would look like across a season, complete with the noise and gaps real feeds actually have, then build a classification and advisory system on top of it.

The system must include:

- A parametric crop-growth-curve simulator covering at least 4 crop types (e.g. rice, wheat, cotton, sugarcane), each generating a realistic vegetation/moisture-index trajectory (emergence → vegetative growth → peak → senescence) driven by documented parameters: planting date, variety/duration class, and a synthetic water-stress driver.
- A noise and cloud-gap injection layer that adds sensor-level noise to the clean simulated curves and drops observations in clustered (not uniformly random) windows, mimicking monsoon-season cloud cover and a realistic satellite revisit cadence.
- A growth-stage ground-truth generator producing a stage label (e.g. sowing, vegetative, flowering, maturity, harvest) at every simulated time-step, derived from the same underlying crop-curve parameters — not inferred from the noisy index.
- A crop-type classifier that ingests only the noisy, gapped simulated index stream (never the clean ground-truth curve) and reports its performance as a full confusion matrix by crop type.
- A growth-stage estimator, evaluated separately from crop-type accuracy, since the two are distinct tasks.

A water-deficit advisory layer that combines the classified growth stage with the simulated moisture-index trajectory to output a decision-support recommendation (e.g. adequate / irrigate within N days / stress detected), calibrated against the simulator's own known synthetic stress driver rather than an invented rule of thumb.

Mandatory Deliverables

1. Working repository with a one-command deploy/run.
2. The parametric crop-growth-curve simulator (≥ 4 crop types) with documented equations and parameters.
3. The noise and cloud-gap injection module, with configurable noise level and gap pattern.
4. The growth-stage ground-truth generator, tied to the same underlying crop-curve parameters used by the simulator.
5. The crop-type classifier with its confusion matrix on held-out simulated data.
6. The growth-stage estimator with its own reported accuracy/error metric.
7. The water-deficit advisory engine with documented decision logic and measured false-alarm/missed-stress rates against simulator ground truth.
8. A live dashboard showing per-plot simulated time series, classified crop and growth stage, and the current advisory, with the persistent simulated-data disclosure banner.
9. A README with an honesty section: what the simulator does and doesn't capture relative to real satellite behavior, and every measured accuracy/error number.

What Is Expected

- A live run generating a fresh simulated season on a new random seed, with the classifier and advisory pipeline operating on it end-to-end.
- The team explaining, on the spot, exactly which parts of the pipeline are simulated versus learned, and why no real dataset was used.
- Transparent reporting of confusion matrices and error rates rather than a single cherry-picked accuracy headline.
- A live demonstration of robustness to noise/gaps — e.g. increasing the gap-dropout rate live and showing how classifier accuracy degrades, proving the system isn't overfit to a clean synthetic case.

Disclaimer

This is a self-contained simulation exercise. All vegetation/moisture-index data, crop-growth curves, and water-stress signals are synthetically generated by the team's own simulator and have not been validated against real satellite imagery, real fields, or real agronomic ground truth. The output must not be presented, marketed, or used as a validated crop-monitoring or irrigation-advisory product, and must not be represented as derived from or validated against ISRO, Bhuvan, or any other real satellite or government dataset.

